

GreenLeaf — Presentation Voice Script (English)

Team Orbit | AI Ideathon 2026

Written to speak directly to the judging rubric: AI Integration (30%), Innovation (25%), User Experience (20%), Technical Execution (15%), Impact & Feasibility (10%). Each section is built so the reasoning behind a feature is stated out loud, not left implicit — judges shouldn't have to infer it.

Slide 1 — Hook

Good [morning/afternoon] everyone. We're **Team Orbit**, and we're here to show you **GreenLeaf** — a multi-agent, AI-powered agrotech advisor built for Bangladeshi farmers.

Here's the problem we're solving: a farmer in Bangladesh makes dozens of high-stakes decisions every season — what to grow, when to plant, how much to spend — almost entirely on instinct and word of mouth. There's no personalized, real-time, trustworthy advisor in their pocket. GreenLeaf is that advisor.

Slide 2 — What GreenLeaf Is

GreenLeaf is an AI-powered multi-agent agricultural advisor. It talks to farmers through natural conversation, pulls live weather data, cross-references trusted agricultural knowledge, and then:

Recommends the right crop for their exact land and conditions

Builds a personalized, dated season plan

Estimates cost and profit before a single taka is spent

Automatically rewrites the plan the moment weather risk changes

And critically — every bit of this happens in **Bangla or English**, whichever the farmer is comfortable in. We didn't bolt language support on afterward; it's core to why this product is usable at all.

Key highlights: monitoring agent · personalized crop prediction · multilingual farmer support · season plan with live profit estimates.

Slide 3 — Architecture & Collaboration (this is where "AI Integration" gets earned)

GreenLeaf isn't one model answering questions — it's a coordinated system of specialized agents orchestrated with **LangGraph**:

A conversation agent that classifies intent and drives the dialogue

A monitoring agent that watches live weather and flags risk

A retrieval agent that pulls from an agronomy knowledge base using RAG

A financial agent that runs deterministic cost, profit, ROI, and break-even calculations

We deliberately split these apart instead of asking one large model to do everything. That's a real architectural decision: it means the financial numbers a farmer sees are never "hallucinated by an LLM" — they come from a deterministic calculation engine, while the LLM handles what it's actually good at, which is conversation and reasoning over retrieved context. That separation is the difference between a chatbot demo and a system a farmer can trust with real money.

Slide 4 — Tech Stack

Under the hood:

Frontend: React 19

Backend: FastAPI

Database: PostgreSQL

Agent orchestration: LangGraph

Reasoning engine: OpenAI API

Live weather: Open-Meteo API

Real-time knowledge fallback: Tavily API

Voice input: Whisper

Plant disease detection: Kindwise crop.health API

Document access: DAM API

Auth: JWT

Deployment: Docker, fully containerized

This isn't a stack chosen to look impressive — every piece maps directly to a farmer-facing capability.

That's what makes this feasible to actually ship, not just demo.

BD Slide 5 — Why This Matters for Bangladesh (Impact & Feasibility)

Agriculture is a massive share of Bangladesh's economy, but most farmers have no access to personalized, real-time, trustworthy advice. Extension officers are stretched thin, online information is fragmented and often wrong, and one bad weather-driven decision can wipe out an entire season's income.

GreenLeaf meets farmers exactly where they are:

Bangla-first, voice-first — literacy is never a barrier

Recommendations grounded in local soil, local weather, local market conditions

Financial planning that respects the farmer's actual budget, not a generic template

Accessible from the same phone already in their pocket — no new hardware, no new habit to build

This isn't a tech demo looking for a use case. It's a use case that's been underserved for decades, finally getting the tool it needs.

Slide 6 — Beyond Bangladesh: The Globalization Story

Here's the question every judge is thinking: does this only work in Bangladesh? No — and that's by design.

GreenLeaf's core architecture is **region-agnostic on purpose**. The agronomy knowledge base, the weather integration, and the financial model are all modular components that can be swapped or extended per region without touching the core agent orchestration. That means:

Other South Asian markets — India, Nepal, Pakistan — can be localized on the same architecture

Smallholder-farming-dominant regions in Southeast Asia and Africa are a natural extension

Multilingual support means adding a new language is a configuration change, not a rebuild

So the framing is: **Bangladesh-first, globally-scalable**. We're not building a Bangladesh tool. We're building the reference implementation of a platform that happens to launch in Bangladesh first, because that's the market with the clearest and most urgent need.

Slides 7–11 — Live Demo: GreenLeaf in Action

Let's show you how this actually works, through five core capabilities.

1 Sequential, Conversational Planning

GreenLeaf doesn't dump a plan on the farmer all at once. It builds one step by step through natural dialogue — land size, soil type, water availability, budget, season — and turns those answers into a logical, dated season plan in real time.

2 Voice or Chat, Farmer's Choice

Farmers can type, or they can just speak. Whisper transcribes voice input, and it's understood and answered exactly the same way as typed text. For a huge portion of our target users, speaking is far more natural than typing — this isn't a nice-to-have, it's an accessibility requirement we treated as core.

3 Diagnose a Crop From a Photo

A farmer snaps a photo of a leaf — say it's turning yellow — and GreenLeaf identifies the likely disease and tells them exactly what steps to take next. This runs through the Kindwise crop.health API, turning a guessing game into a diagnosis.

4 Dynamic Financial Re-Planning — the standout feature

This is where GreenLeaf earns its keep. If a farmer says their budget just dropped 10%, or 20%, GreenLeaf doesn't just scale everything down proportionally — it recalculates the entire plan to **maximize achievable revenue within the new constraint**. Cost, profit, ROI, and break-even are all recomputed deterministically, live, in the conversation. This is the feature that turns GreenLeaf from an advisor into a financial co-pilot.

5 Never a Dead End: RAG + Web Search Fallback

For general farming questions, GreenLeaf first searches its own agronomy knowledge base using RAG. If it doesn't find an answer there, it automatically falls back to a live web search and surfaces that instead. A farmer never hits "I don't know" — the system always finds a path to an answer.

Slide 12 — Roadmap: Where This Goes Next

We're not stopping at the hackathon build.

Near-term:

- Real-time market price integration, so farmers know which crop is actually most profitable *right now*
- SMS/IVR fallback for zero-connectivity areas
- Offline-first mobile app for low-connectivity regions

Mid-term:

- Satellite imagery for field-level crop health monitoring
- A community layer where farmers share real outcomes with each other
- Integration with government subsidy and crop insurance schemes

Long-term / Globalization:

- Region-specific, plug-and-play agronomy knowledge modules
- Language expansion — Hindi, Nepali, Swahili, and beyond
- Predictive analytics that forecast next season's outcomes from historical data

Technical roadmap:

- Break the current LangGraph architecture into independently scalable microservices per agent

Partition the vector database by region for faster, more precise retrieval

Add edge caching to cut latency on weather and market data

Move to fine-tuned smaller models for cost-efficient inference at scale, keeping unit economics sane as we grow

Slide 13 — Close

GreenLeaf isn't just a hackathon project. It's a vision where every farmer, from the phone already in their pocket, gets advice on par with an expert agronomist. We're starting in Bangladesh — because that's where the need is sharpest — but we built this to go global from day one.

Thank you. We're Team Orbit.